

Constructing Parallel and Perpendicular Lines

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Perpendicular Lines

Perpendicular lines are lines that intersect at a right angle (90°).

Parallel Lines

Parallel lines are lines in the same plane that never intersect, no matter how far they are extended.

Perpendicular Through a Point

To construct a **perpendicular line** through a point on a line:

Use a compass to mark two points equidistant from the given point.

From these two points, draw arcs with equal radii that intersect above and below the line.

Connect the intersection points to form the perpendicular line.

Perpendicular Through an External Point

To construct a **perpendicular line** through a point not on a line:

Use a compass centered at the given point to create arcs intersecting the line at two points.

From these intersection points, draw arcs below the line.

Connect the given point to the arc intersection to form the perpendicular.

Constructing a Parallel Line

To construct a **parallel line** through a given point:

Draw a transversal through the given point intersecting the original line.

Copy the angle formed using a compass and straightedge.

Use the **corresponding angles theorem** or **alternate interior angles theorem**.

Essential Tools

The essential tools for geometric constructions are:

Compass

Straightedge

Pencil

Examples

Example 1

Given line AB and point P on the line, construct a perpendicular line to AB passing through point P .

Example 2

Point Q is located 5 cm away from line CD .
Construct a perpendicular line from point Q to line CD .

Example 3

Given line m and point R not on the line, construct a line parallel to m passing through point R .

Example 4

Construct a square with side length 6 cm using only perpendicular line constructions.

Example 5

Given a line segment EF of length 8 cm, construct its perpendicular bisector.

Practice Exercises

Practice 1

Given line AB and point P on the line, construct a perpendicular line to AB passing through point P .

Practice 2

Point Q is located 5 cm away from line CD .
Construct a perpendicular line from point Q to line CD .

Practice 3

Given line m and point R not on the line, construct a line parallel to m passing through point R .

Practice 4

Construct a square with side length 6 cm using only perpendicular line constructions.

Practice 5

Given a line segment EF of length 8 cm, construct its perpendicular bisector.

Activities

Activity 1

Point N is located 3 cm away from line ST .
Construct a perpendicular line from point N to line ST .

Activity 2

Given line j and point W not on the line, construct a line parallel to j passing through point W .

Activity 3

Construct a square with side length 3 cm using only perpendicular line constructions.

Activity 4

Given a line segment GH of length 4 cm, construct its perpendicular bisector.

Thank you!